EN3160 - Assignment 2

Fitting and Alignment

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1. Blob Detection: Sunflower Field

Blob detection was implemented using a scale-normalized Laplacian of Gaussian (LoG) filter and scale-space extrema detection. A 3D scale-space volume was generated across multiple σ values, and local maxima were extracted using a 3D maximum filter.

```
        # Core LoG and Extrema Logic
        for sigma in sigmas:
        blurred = cv.GaussianBlur(gray, (ksize, ksize), sigma)
        normalized_log = cv.Laplacian(blurred, cv.CV_64F, ksize=3) * (sigma ** 2)
        log_responses.append(normalized_log)

        log_volume = np.abs(np.stack(log_responses, axis=-1))
        blob_mask = (log_volume == maximum_filter(log_volume, size=7)) & (log_volume > 0.4 * np
    .max(log_volume))

```



Figure 1: Detected sunflowers

The threshold multiplier (0.4) and 3D neighborhood size (7) were tuned to filter out background noise like clouds and trees. The algorithm successfully locked onto the high-contrast central seed disks of the sunflowers rather than the outer petals.

Scale range: 6.10 to 7.63.

Largest circles parameters: Rank 1: Center = (0, 275), Radius = 10, $\sigma = 7.63$ — Rank 2: Center = (282, 338), Radius = 10, $\sigma = 7.63$ — Rank 3: Center = (106, 255), Radius = 8, $\sigma = 6.10$.

2. Robust Model Fitting (RANSAC)

Methodology and Code

RANSAC was implemented to sequentially extract models from noisy data. For line estimation, two points were iteratively sampled to compute the normal $[a, b]^T$, enforcing $||[a, b]^T|| = 1$. The normal distance error threshold was carefully selected as 1.5, and a model was only accepted if it met a minimum consensus threshold of 15 points. Line inliers were identified and removed from the dataset. RANSAC then sampled three points from the remnant to algebraically fit a circle, utilizing a radial error threshold of 1.5 and a minimum consensus requirement of 15 points. Finally, the line was re-fit using Total Least Squares, and the circle was optimized using `scipy.minimize`.

```
        # Normal distance for line consensus
        a, b = normal / np.linalg.norm(normal)
        distances_line = np.abs(X[:, 0]*a + X[:, 1]*b + d)
        inliers_line = np.where(distances_line < 1.5)[0]

        # Radial error for circle consensus
        distances_circ = np.sqrt((X_remnant[:, 0] - cx)**2 + (X_remnant[:, 1] - cy)**2)
        errors_circ = np.abs(distances_circ - r_est)
        inliers_circ = np.where(errors_circ < 1.5)[0]

```

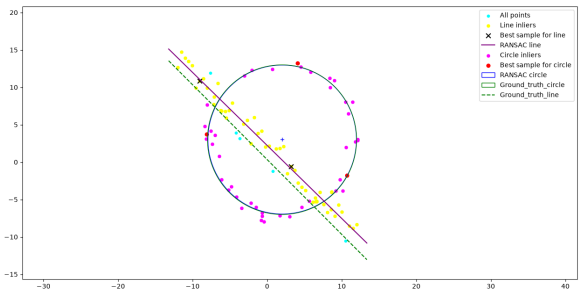


Figure 2: Sequential RANSAC fitting. Line inliers (yellow) were removed before estimating the circle inliers (magenta).

Discussion

What will happen if we fit the circle first?

If RANSAC fits the circle prior to the line, it aims to maximize the consensus set across the entire point cloud. The circle model may over-fit by erroneously claiming noisy points that actually belong to the intersecting line. When this flawed consensus set is subtracted, the remnant will contain a fragmented, incomplete set of line points, causing the subsequent line estimation to skew heavily or fail entirely.